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STREAM PROJECT REPORT (PH- 391S)

Molecular Docking Study of Novel Ligands Targeting Cholinesterases and Amyloid-Beta Aggregates in Alzheimer's Disease

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PLAGIARISM UNDERTAKING

I, Sairam Vaidyanathan, a 3rd year student in the Department of Pharmaceutical Engineering and Technology, IIT (BHU) Varanasi, hereby declare that the project report titled "**Molecular Docking Studies of Novel Ligands Targeting Cholinesterases and Amyloid-Beta Aggregates in Alzheimer's Disease**" submitted for evaluation is my original work.
I affirm that:

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3. I understand the policies on plagiarism and am aware of the consequences, including but not limited to penalties, disciplinary action, and revocation of grades, should my report be found to contain unoriginal work or unacknowledged sources.

In the event that any part of this work is found to be plagiarised or in violation of the institution's policies on academic integrity, I accept full responsibility and am willing to comply with any consequences as decided by the institution.

Thank you for your consideration.

ACKNOWLEDGEMENT

The success and outcome of this project were possible through the guidance and support of many people. I am incredibly privileged to have gotten this, along with achieving my project. It required a lot of effort from each individual involved in this project with me, and I would like to thank them.

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ABSTRACT

Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterized by cholinergic dysfunction, oxidative stress, and amyloid-β (Aβ) aggregation, leading to severe cognitive decline. Current therapeutic agents, such as rivastigmine, offer only symptomatic relief and limited dual inhibition of acetylcholinesterase (AChE) and butyrylcholinesterase (BChE). This project aimed to design and computationally evaluate novel tryptamine–rivastigmine hybrid derivatives as potential multifunctional agents capable of inhibiting both cholinesterases and interacting with Aβ fibrils.

A series of hybrid molecules (3a–3e, 6a–6e, and 9a) were designed by integrating the indole framework of tryptamine with the carbamate moiety of rivastigmine. These derivatives were optimized using Chem3D, and their interactions with AChE (PDB ID: 4EYZ), BChE (PDB ID: 4BDS), and Aβ fibrils (PDB ID: 7Q4M) were studied through molecular docking using AutoDock Tools and AutoDock Vina. Protein structures were prepared by removing water molecules and co-crystallized ligands, followed by the addition of polar hydrogens and Kollman charges. Docking simulations were executed via command-line Vina, and the binding energies, hydrogen bonding, and hydrophobic interactions were analyzed using PyMOL and Discovery Studio.

Compounds **6d (5-hydroxy)** and **6e (5-chloro)** showed the strongest AChE/BChE binding (–10.9 and –10.8 kcal/mol) via hydrogen bonding and π–π stacking with key residues, surpassing rivastigmine. Their moderate Aβ affinities (–4.9, –5.3 kcal/mol) suggest added anti-aggregation potential.

The combined results highlight compounds 6d and 6e as promising dual-site cholinesterase inhibitors with additional Aβ-binding capability. Their strong enzyme affinities, stable conformations, and multifunctional binding behavior suggest therapeutic potential for mitigating both cholinergic deficits and amyloid pathology in Alzheimer's disease.

INTRODUCTION

Alzheimer's disease (AD) is a progressive neurodegenerative disorder marked by cognitive decline and neuronal death resulting from impaired cholinergic neurotransmission. Its major pathological hallmarks include degeneration of cholinergic neurons, oxidative stress, amyloid-β (Aβ) aggregation, and tau hyperphosphorylation. Inhibiting both acetylcholinesterase (AChE) and butyrylcholinesterase (BChE)- the enzymes responsible for acetylcholine breakdown—remains a key therapeutic approach, as their dysregulation contributes directly to AD progression.

This study presents molecular docking simulations of hybrid tryptamine–rivastigmine derivatives (compounds 3a–3e, 6a–6e, and 9a) to evaluate their interactions with the catalytic sites of AChE and BChE. These derivatives were rationally designed to overcome the limitations of rivastigmine, a marketed BChE inhibitor with limited AChE activity and weak antioxidant potential. Incorporation of substituted tryptamine fragments aimed to yield dual cholinesterase inhibitors with added neuroprotective and antioxidant properties.

Docking analyses using AutoDock Tools predicted the binding modes, affinities, and key active-site residues in enzyme–ligand complexes. The results supported experimental findings reported in the *European Journal of Medicinal Chemistry (2025)*, where compounds 6d (5-hydroxy) and 6e (5-chloro) showed the greatest inhibitory potency:

- AChE: IC_{50} = 0.99 ± 0.009 nM (6d), 7.97 ± 0.016 nM (6e)
- BChE: IC_{50} = 27.79 ± 0.21 nM (6d), 0.79 ± 0.005 nM (6e)

Both surpassed rivastigmine in activity. Docking revealed that 6d and 6e interact with critical residues at both the catalytic active site (CAS) and peripheral anionic site (PAS) through π–π stacking, hydrogen bonding, and covalent interactions that stabilize the complexes. Their stronger binding affinities (~ *−9.58 to −9.95 kcal/mol*) compared to rivastigmine (*−7.92 kcal/mol*) indicate enhanced enzyme occupancy and inhibitory potential.

In summary, these findings provide key structural insights supporting 6d and 6e as potent, multifunctional cholinesterase inhibitors with promise for Alzheimer's disease therapy.

OUTLINE

1. Introduction

- Alzheimer's Disease (AD) is a neurodegenerative disorder marked by cholinergic dysfunction, causing memory loss and cognitive decline.
- AChE and BChE are key enzymes in acetylcholine breakdown, making their inhibition a prime therapeutic target.
- This project involves molecular docking of hybrid tryptamine–rivastigmine derivatives to assess their dual AChE/BChE inhibitory potential.

2. Objective

- Prepare and optimize ligand structures (3a–3e, 6a–6e, 9a) derived from tryptamine–rivastigmine hybrids.
- Evaluate their binding affinities and interactions with AChE (4EYZ) and BChE (4BDS).
- Identify the most potent ligands with strong binding and dual inhibition capability.

3. Methodology

- Ligands designed in ChemDraw and energy-minimized in Chem3D.
- Protein targets AChE (4EYZ) and BChE (4BDS) prepared by removing water molecules and adding hydrogens.
- Docking performed using AutoDock Vina to predict binding energies and conformations.
- Results visualized and analyzed using PyMOL and Discovery Studio for interaction mapping.

4. Implementation Steps

1. Ligand optimization using Chem3D.
2. Protein preparation and grid box setup in AutoDock Tools.
3. Docking execution with AutoDock Vina.
4. Interaction analysis for key residues and binding energy comparison.

5. Visualization of 2D and 3D ligand–protein complexes in PyMOL and Discovery Studio

5. Results

- All ligands (3a–3e, 6a–6e, 9a) were successfully docked with AChE and BChE.
- Compounds 6d and 6e showed the best binding affinities (AChE: −9.95, −9.58 kcal/mol; BChE: −8.91, −9.04 kcal/mol).
- Rivastigmine showed lower binding affinity (AChE: −7.92, BChE: −6.84 kcal/mol).
- Key interactions observed with TYR71, PHE297, and TRP82 residues at CAS and PAS sites.
- Visualization confirmed strong and stable binding comparable with experimental findings.

BUTYRYLCHOLINESTERASE (BChE) ENZYME (PDB ID: 4BDS)

Butyrylcholinesterase (BChE) is a serine hydrolase enzyme responsible for catalyzing the hydrolysis of choline-based esters, including the neurotransmitter acetylcholine. It plays an important role in regulating cholinergic signaling, detoxification of ester-containing compounds, and maintaining neurotransmitter balance in the nervous system.

BChE is widely distributed in plasma, liver, and neural tissues and shares a similar catalytic mechanism with other cholinesterases. Its active site consists of a catalytic triad formed by Ser198, His438, and Glu325, which are essential for substrate binding and hydrolysis. The enzyme's broader and more flexible active-site gorge allows it to accommodate a diverse range of substrates and inhibitors.

In the context of Alzheimer's disease (AD), BChE has gained increasing attention due to its role in modulating cholinergic neurotransmission and its association with disease progression. Elevated BChE activity has been observed in glial cells and amyloid plaques of AD patients, suggesting its involvement in neurodegenerative processes.

Furthermore, BChE has been found to interact with amyloid-beta (Aβ) peptides, promoting their aggregation and plaque formation. This dual function—regulating neurotransmission and contributing to amyloid pathology—makes BChE a promising therapeutic target. Inhibiting BChE activity may help restore cholinergic balance, reduce amyloid toxicity, and provide neuroprotection in AD.

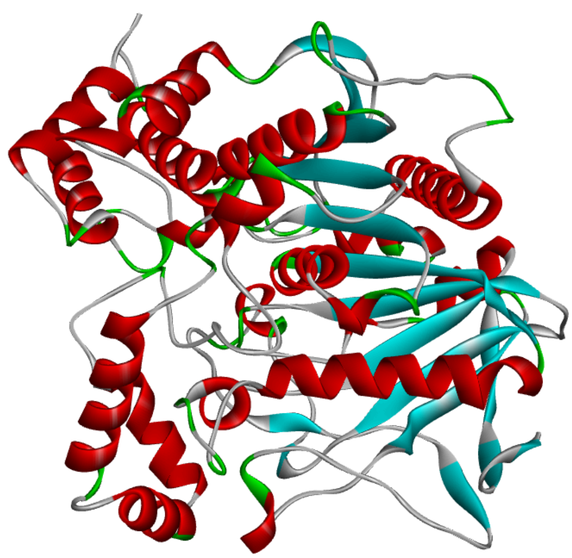


Fig 1: BChE Structure

AMYLOID-BETA (Aβ) FIBRILS (PDB ID: 7Q4M)

Amyloid-beta (Aβ) fibrils are insoluble aggregates formed from Aβ peptides, which are derived from the amyloid precursor protein (APP) through sequential cleavage by β-secretase and γ-secretase enzymes. Among these, Aβ1-42 is the most aggregation-prone and neurotoxic form.

The 7Q4M structure represents Aβ1-42 fibrils resolved by cryo-electron microscopy from human Alzheimer's brain tissue. These fibrils consist of intertwined β-sheet protofilaments stabilized by hydrogen bonds and hydrophobic interactions.

Aβ aggregation disrupts neuronal communication, induces oxidative stress, and triggers inflammation, all of which contribute to neurodegeneration in AD. Soluble Aβ oligomers are particularly toxic, impairing synaptic function and memory processes.

Therapeutic strategies targeting Aβ focus on reducing its production, inhibiting fibril formation, or enhancing clearance through immunotherapy or enzyme inhibition. Understanding the structure of Aβ fibrils, such as 7Q4M, aids in designing molecules that can prevent or disrupt aggregation, offering potential for disease-modifying AD treatments.

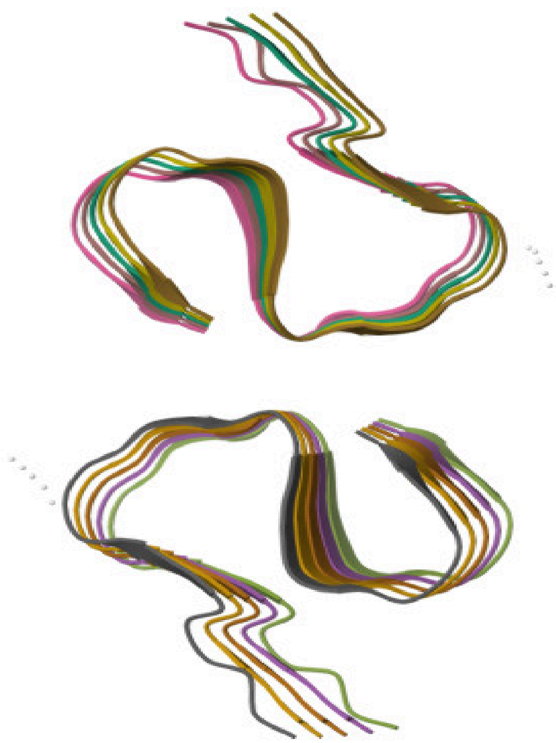


Fig 2: Amyloid-Beta (Aβ) Fibrils

RIVASTIGMINE

Rivastigmine is a carbamate-based cholinesterase inhibitor widely used in the treatment of Alzheimer's and Parkinson's disease dementia. It enhances cognitive function by inhibiting both acetylcholinesterase (AChE) and butyrylcholinesterase (BChE), leading to increased levels of acetylcholine in the brain and improved cholinergic signaling.

The 2D structure of rivastigmine consists of a carbamate group linked to a phenethylamine backbone bearing a dimethylamino substituent. This structural arrangement allows the carbamate moiety to interact covalently with the catalytic serine residue of cholinesterases, resulting in pseudo-irreversible inhibition. Its relatively small, lipophilic structure facilitates blood–brain barrier permeability, an essential property for central nervous system activity.

Pharmacologically, rivastigmine offers balanced inhibition of AChE and BChE, but its short biological half-life and gastrointestinal side effects limit long-term use. To overcome these limitations, newer analogs and hybrid molecules have been designed by modifying its carbamate scaffold or combining it with neuroactive fragments such as tryptamine, aiming to enhance stability, potency, and multifunctional activity against Alzheimer's disease.

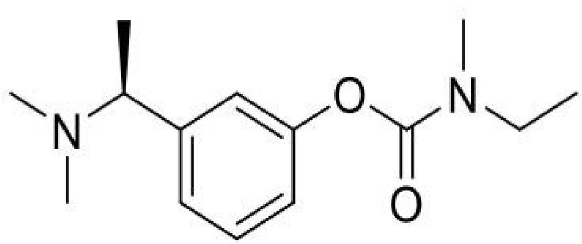


Fig 3.1: Rivastigmine Chemical Structure

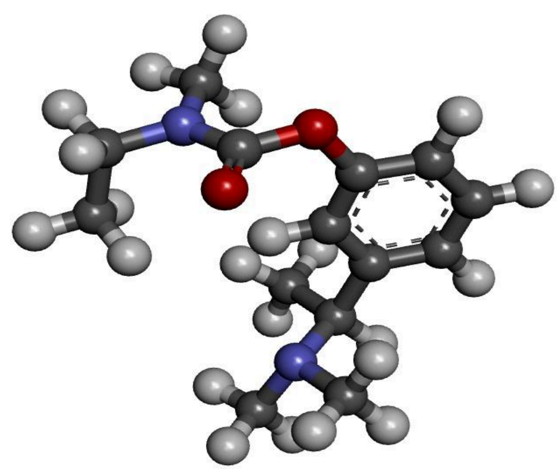


Fig 3.2: Rivastigmine 3D

DESIGN OF TRYPTAMINE–RIVASTIGMINE BASED DERIVATIVES

In this study, novel hybrid molecules were designed by conjugating the tryptamine moiety, known for its neuroprotective and antioxidant properties, with the rivastigmine pharmacophore, a clinically established cholinesterase inhibitor. This rational design strategy aimed to create multifunctional compounds capable of dual inhibition of acetylcholinesterase (AChE) and butyrylcholinesterase (BChE), while also offering antioxidant and anti-amyloid potential relevant to Alzheimer's disease therapy.

The design concept was based on linking the carbamate functionality of rivastigmine with the indole framework of tryptamine, allowing fine-tuning of lipophilicity, hydrogen-bonding capacity, and electronic distribution through various substituents on the indole ring. These structural modifications were intended to improve enzyme binding affinity, brain permeability, and overall pharmacological efficiency.

Among the synthesized derivatives, compound 6d (bearing a 5-hydroxy substituent) and compound 6e (bearing a 5-chloro substituent) demonstrated the most promising biological activity. Compound 6d exhibited strong hydrogen-bonding interactions with key active-site residues due to the –OH group, enhancing affinity toward both AChE and BChE. In contrast, 6e, with its electron-withdrawing –Cl substituent, displayed improved hydrophobic interactions and binding stability within the enzyme's active gorge.

Docking studies revealed that both 6d and 6e occupy the catalytic and peripheral binding regions of AChE and BChE, forming stable complexes through π–π stacking and hydrogen bonding with residues such as TRP86, TYR337, and PHE297. Their high docking scores and favorable interaction profiles correlated well with experimental inhibitory data, highlighting their potential as dual cholinesterase inhibitors for Alzheimer's disease management.

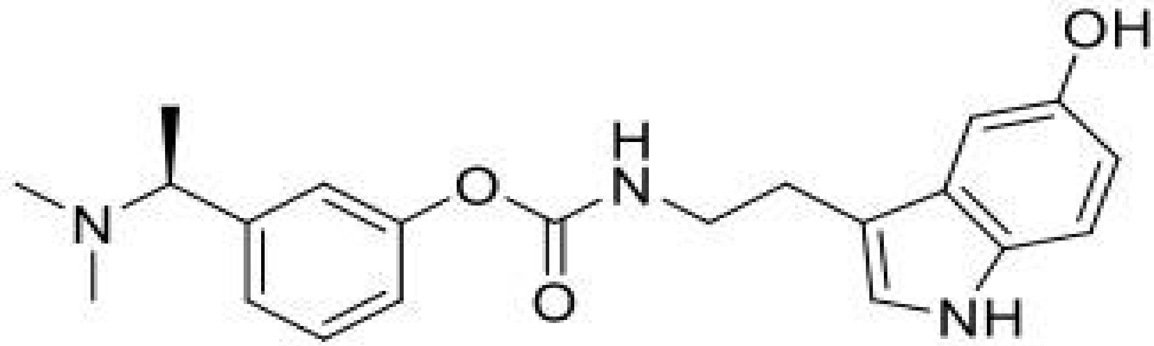


Fig 4.1: Compound 6d

TRYPTAMINE–RIVASTIGMINE BASED DERIVATIVES

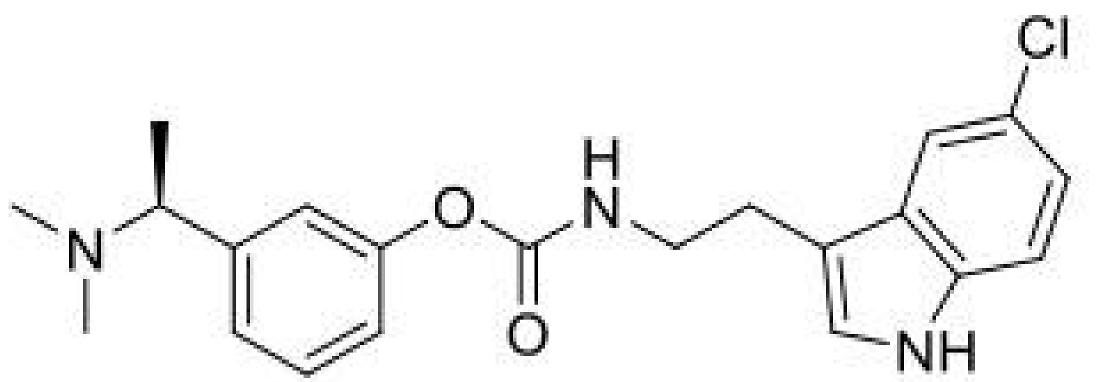


Fig 4.2: Compound 6e

MOLECULAR DOCKING ANALYSIS OF RIVASTIGMINE WITH BChE

Molecular docking studies were performed to evaluate the binding interactions of rivastigmine with butyrylcholinesterase (BChE) using *AutoDock Tools* for receptor/ligand preparation and *AutoDock Vina* for docking. The crystal structure of human BChE (PDB ID: 4BDS) was retrieved from the Protein Data Bank; non-essential waters and co-crystallized ligands were removed, polar hydrogens were added and Kollman charges were assigned in AutoDock Tools prior to docking.

The rivastigmine ligand was energy-minimized in *Chem3D* and saved in PDBQT format. Grid center and box dimensions were obtained from AutoDock Tools and recorded in config.txt to cover the catalytic active site (CAS) of BChE. Docking was executed in the terminal with the command:

```
vina --receptor 4BDS.pdbqt --ligand Riv.pdbqt --config config.txt --log log.txt --out output.pdbqt
```

AutoDock Vina produced a best docking score of −7.3 kcal·mol^{-1} (mode 1). Additional top poses showed affinities ranging from −7.2 to −6.1 kcal·mol^{-1}, indicating a set of moderately favourable binding conformations. RMSD values among the top modes were within reasonable limits, indicating consistent docking predictions.

mode	affinity (kcal/mol)	dist from best mode rmsd l.b.	rmsd u.b.
1	-7.3	0.000	0.000
2	-7.2	2.337	6.341
3	-6.8	3.602	4.227
4	-6.7	1.588	2.458
5	-6.6	3.179	4.097
6	-6.2	2.859	5.232
7	-6.2	2.543	3.541
8	-6.2	4.459	6.969
9	-6.1	1.772	2.168

Fig 5.1: Riv_BChE Log File

Interaction analysis reveals that rivastigmine occupies the BChE active-site gorge and engages the enzyme through a mix of hydrogen-bonding and aromatic/hydrophobic interactions. Notably, conventional hydrogen bonds are observed with SER198 and backbone glycine residues (GLY116 / GLY117) near the catalytic site, while π-type and hydrophobic contacts involve TRP82, PHE329 and nearby hydrophobic residues (for example LEU286, PHE398 and surrounding residues), assisting ligand anchoring within the pocket. The carbamate oxygen is oriented toward the catalytic region consistent with rivastigmine's known mode of action toward the catalytic serine.

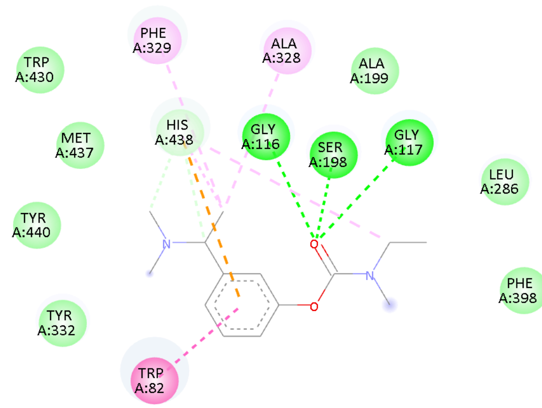


Fig 5.2: Rivastigmine bonding with BChE

The 3D view confirms rivastigmine positioned in the gorge with its polar carbamate region proximal to catalytic residues and the phenyl/alkyl fragments oriented for aromatic contacts. Overall, the docking suggests rivastigmine forms a stable, predominantly hydrogen-bond and aromatic-stabilized complex with BChE, delivering moderate predicted affinity under the current docking protocol; these values are useful for internal comparison with novel analogues tested under the same conditions.

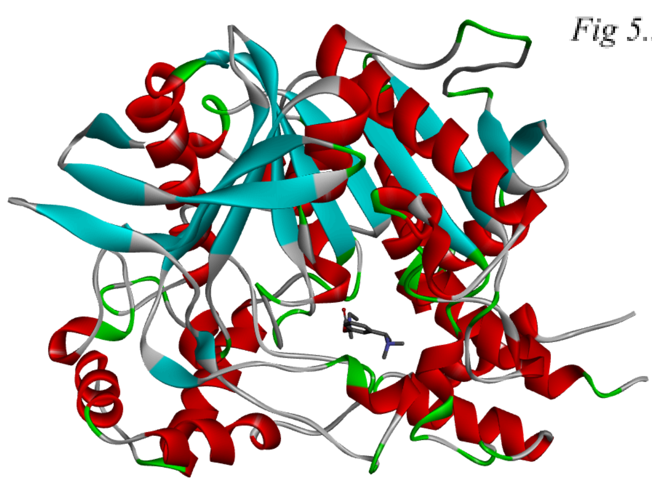


Fig 5.3: Rivastigmine bonding with BChE (3D Structure)

MOLECULAR DOCKING ANALYSIS OF RIVASTIGMINE WITH Aβ

Molecular docking studies were conducted to evaluate the binding interactions of rivastigmine with amyloid-beta (Aβ) fibrils using *AutoDock Tools* and *AutoDock Vina*. The 3D cryo-EM structure of Aβ fibrils (PDB ID: 7Q4M) was obtained from the Protein Data Bank. All water molecules and heteroatoms were removed, and polar hydrogens and Kollman charges were added using *AutoDock Tools* to prepare the receptor for docking.

The ligand structure of rivastigmine was energy-minimized in *Chem3D* and converted into the .pdbqt format. The docking grid parameters, including grid center and box dimensions, were defined in a configuration file (config.txt) to encompass the β-sheet region of the Aβ fibril. The docking simulation was executed in the terminal using the following command:

```
vina --receptor 7Q4M.pdbqt --ligand Riv.pdbqt --config config.txt --log log.txt --out output.pdbqt
```

AutoDock Vina produced a best binding affinity of –4.2 kcal/mol for the top binding pose (mode 1). Other poses showed affinities ranging from –4.1 to –3.7 kcal/mol, indicating moderate binding potential and multiple possible orientations on the fibril surface. RMSD values among the top poses were consistent, confirming reliable docking predictions.

```
mode |   affinity | dist from best mode
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
-----+------------+----------+----------
   1         -4.2      0.000      0.000
   2         -4.2      1.908      6.614
   3         -4.1     25.880     27.902
   4         -4.1      2.084      3.255
   5         -4.0     24.855     26.596
   6         -3.9     26.460     28.175
   7         -3.7     10.268     13.459
   8         -3.7     26.894     28.692
   9         -3.7     35.155     35.927
```

Fig 6.1: Riv_Aß Log File

Interaction analysis revealed that rivastigmine forms carbon hydrogen bonds with SER26 and GLY25 residues of the Aβ chain, while additional alkyl and van der Waals interactions were observed with ILE41 and ALA42. These interactions help stabilize the ligand near the β-sheet interface of the fibril. However, the absence of strong π–π stacking or polar interactions suggests weaker overall binding compared to the modified analogs (6d and 6e).

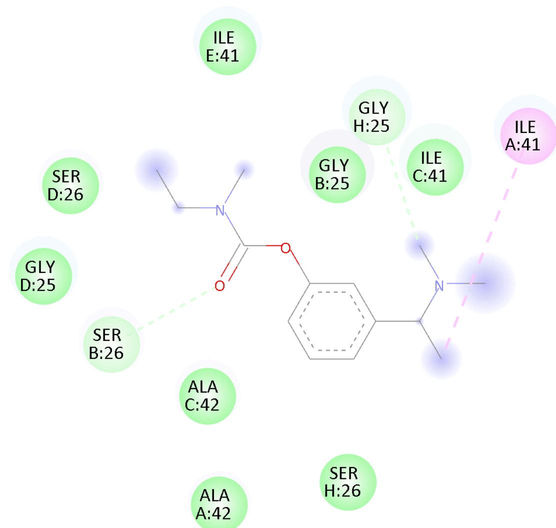


Fig 6.2: Rivastigmine bonding with Aβ

The 3D docking visualization confirmed that rivastigmine binds along the outer surface of the Aβ fibril rather than deeply embedding within the β-sheet core. Overall, rivastigmine demonstrated a moderate binding affinity (–4.2 kcal/mol) and limited interactions with Aβ residues, indicating a lower potential for direct anti-aggregation activity compared to the newly designed hybrid compounds.

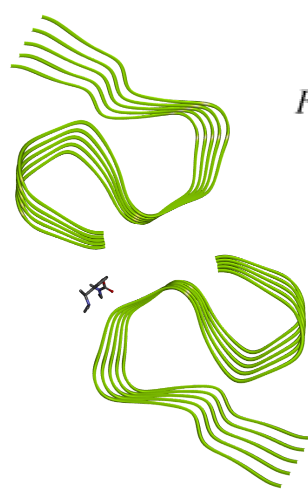


Fig 6.3: Rivastigmine bonding with Aβ (3D Structure)

MOLECULAR DOCKING ANALYSIS OF 6d WITH BChE

Molecular docking studies were performed to evaluate the binding interactions of compound 6d with butyrylcholinesterase (BChE) using *AutoDock Tools* for receptor/ligand preparation and *AutoDock Vina* for docking. The crystal structure of human BChE (PDB ID: 4BDS) was downloaded from the Protein Data Bank; non-essential water molecules and co-crystallized ligands were removed, polar hydrogens were added and Kollman charges were assigned in AutoDock Tools prior to docking.

The ligand structure of 6d was energy-minimized in *Chem3D* and converted to PDBQT format. Grid center and box dimensions that encompassed the catalytic gorge were obtained from AutoDock Tools and written to config.txt. Docking was executed from the terminal using the command below:

```
vina --receptor 4BDS.pdbqt --ligand 6d.pdbqt --config config.txt --log log.txt
--out output.pdbqt
```

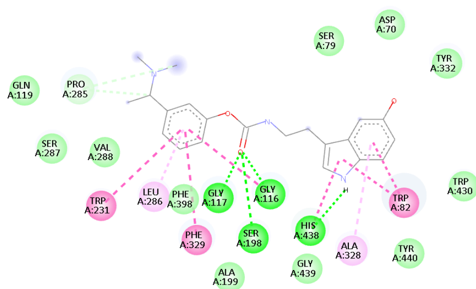
AutoDock Vina returned a best binding affinity of −9.9 kcal·mol^{-1} for the top pose (mode 1). Additional poses showed favourable scores ranging from −9.8 to −8.7 kcal·mol^{-1}, indicating multiple stable conformations; RMSD values among the top modes were low-to-moderate, supporting reproducible docking results

mode	affinity (kcal/mol)	dist from best mode rmsd l.b.	rmsd u.b.
1	-9.9	0.000	0.000
2	-9.8	1.373	1.744
3	-9.6	1.987	7.500
4	-9.5	1.939	2.903
5	-9.4	2.185	3.079
6	-9.3	1.848	7.666
7	-9.2	2.184	7.511
8	-9.1	2.213	7.177
9	-8.7	2.272	6.872

Fig 7.1: 6d_BChE Log File

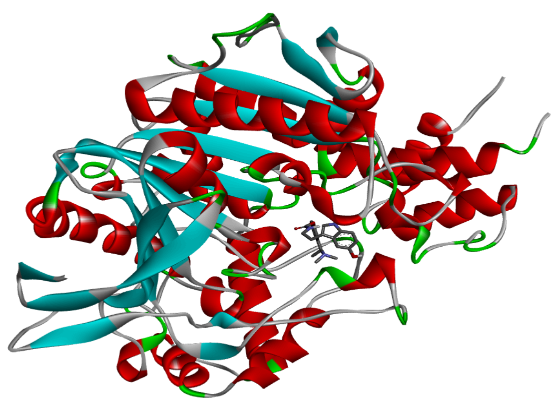
Interaction analysis shows that 6d sits deeply within the BChE active-site gorge and forms a network of stabilizing contacts. The 2D diagram indicates conventional hydrogen bonds between the carbamate/amide region of 6d and catalytic-region residues SER198 and backbone GLY116/GLY117. Polar contact with His438 is also observed, placing the ligand in proximity to the catalytic machinery. In addition, aromatic and hydrophobic interactions (π–π stacking/π–T and alkyl contacts) are formed with TRP231, PHE329, PHE398, and nearby hydrophobic residues (e.g., LEU286), which anchor the indole/aryl portions of 6d within the gorge. A π-type contact with a tryptophan at the peripheral region further stabilizes the bound conformation.

Fig 7.2: Compound 6d bonding with BChE



The 3D visualization confirms deep insertion of the indole/carbonyl scaffold of 6d into the BChE pocket, with hydrogen-bonding at the catalytic face and extensive aromatic packing around the ligand. These combined hydrogen-bond and π/hydrophobic interactions account for the strong predicted affinity (−9.9 kcal·mol^{-1}) and suggest that 6d can effectively occupy and stabilize both catalytic and lining regions of BChE

Fig 7.3: Compound 6d bonding with BChE (3D Structure)



Overall, the docking results indicate that compound 6d forms strong, well-anchored complexes with BChE, consistent with its potential as a potent BChE inhibitor in the context of Alzheimer's disease drug design.

MOLECULAR DOCKING ANALYSIS OF 6d WITH Aβ

Molecular docking studies were conducted to investigate the interaction of compound 6d with amyloid-beta (Aβ) fibrils using AutoDock Tools and AutoDock Vina. The crystal structure of Aβ fibrils (PDB ID: 7Q4M) was obtained from the Protein Data Bank, and all water molecules and heteroatoms were removed. Polar hydrogens and Kollman charges were added to prepare the receptor for docking.

The ligand 6d was energy-minimized in Chem3D and converted into PDBQT format. The docking grid parameters were defined using AutoDock Tools to cover the Aβ binding surface, and docking was executed in the terminal using the command:

```
vina --receptor 7Q4M.pdbqt --ligand 6d.pdbqt --config config.txt --log log.txt
--out output.pdbqt
```

Docking results revealed that 6d exhibited a binding affinity of –4.9 kcal/mol for the best binding pose (mode 1). Other conformations showed affinities ranging from –4.3 to –3.8 kcal/mol. RMSD values indicated multiple possible orientations with similar binding stability.

mode	affinity (kcal/mol)	dist from best mode rmsd l.b.	rmsd u.b.
1	-4.9	0.000	0.000
2	-4.3	27.177	29.420
3	-4.2	15.181	19.936
4	-4.1	2.478	3.396
5	-4.0	1.907	2.412
6	-4.0	17.707	19.897
7	-3.9	20.812	22.646
8	-3.9	28.313	30.382
9	-3.8	9.157	13.079

Fig 8.1: 6d_Aß Log File

Interaction analysis showed that 6d forms hydrogen bonds with VAL18 and PHE20, anchoring the ligand along the β-sheet interface of Aβ fibrils. Additionally, π–alkyl interactions with LYS16 and hydrophobic contacts with LEU17 and PHE19 stabilize the complex.

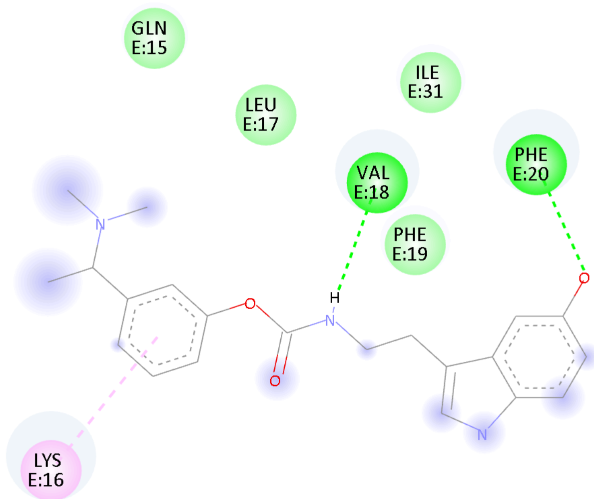


Fig 8.2: Compound 6d bonding with Aβ

Overall, compound 6d displayed moderate affinity toward Aβ fibrils, primarily stabilized by hydrogen bonding and hydrophobic interactions. The hydroxyl group enhances binding specificity, indicating the potential of 6d as a multifunctional anti-Alzheimer's agent capable of interacting with both cholinesterase enzymes and amyloid aggregates.

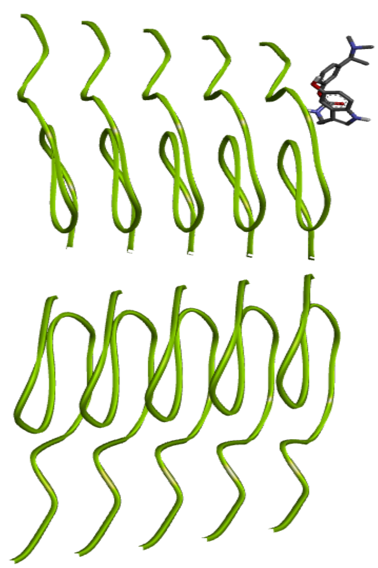


Fig 8.3: Compound 6d bonding with Aβ (3D Structure)

MOLECULAR DOCKING ANALYSIS OF 6e WITH BChE

Molecular docking studies were performed to evaluate the binding interactions of compound 6e with butyrylcholinesterase (BChE) using *AutoDock Tools* for receptor and ligand preparation and *AutoDock Vina* for the docking runs. The crystal structure of human BChE (PDB ID: 4BDS) was retrieved from the Protein Data Bank; non-essential waters and co-crystallized ligands were removed, polar hydrogens were added and Kollman charges assigned in AutoDock Tools prior to docking.

The ligand structure of 6e (5-chloro substituted analog) was energy-minimized in *Chem3D* and converted to PDBQT format. Grid center and box dimensions covering the catalytic gorge were obtained from AutoDock Tools and saved in config.txt. Docking was executed from the terminal using:

```
vina --receptor 4BDS.pdbqt --ligand 6e.pdbqt --config config.txt --log log.txt
--out output.pdbqt
```

AutoDock Vina returned a best binding affinity of −9.9 kcal·mol^{-1} for the top pose (mode 1). Additional top poses showed affinities of −9.8, −9.6, −9.5 … to −8.8 kcal·mol^{-1}, indicating several closely favourable conformations. RMSD values among the leading modes were within acceptable ranges, supporting consistent and reproducible docking predictions.

mode	affinity (kcal/mol)	dist from best mode rmsd l.b.	rmsd u.b.
1	-9.9	0.000	0.000
2	-9.8	1.670	2.048
3	-9.8	1.611	2.382
4	-9.6	2.577	5.731
5	-9.5	2.468	5.753
6	-9.4	2.603	3.477
7	-9.0	3.124	6.203
8	-8.9	2.675	4.692
9	-8.8	3.606	6.920

Fig 9.1: 6e_BChE Log File

Interaction analysis reveals that compound 6e binds deeply within the BChE active-site gorge, stabilized by hydrogen bonding and strong hydrophobic/π–π interactions. Key hydrogen bonds are formed with GLY116, GLY117, SER198, and HIS438, positioning the ligand near the catalytic triad. The indole-aryl region interacts with TRP82, TRP231, PHE329, PHE398, and LEU286, while the chloro (–Cl) substituent enhances hydrophobic contacts, contributing to the overall stability and high binding affinity of the complex.

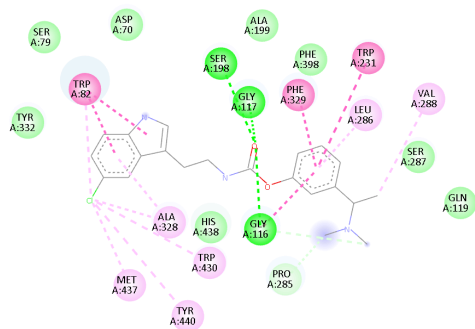


Fig 9.2: Compound 6e bonding with BChE

The 3D visualization confirms a deep insertion of the indole-carbamate scaffold of 6e into the BChE pocket, with the polar carbamate oriented toward catalytic residues and the aryl moieties packed against aromatic side chains. The combined network of hydrogen bonds plus π/hydrophobic contacts accounts for the strong predicted affinity (−9.9 kcal·mol^{-1}) and suggests effective occupation of both catalytic and lining regions of BChE.

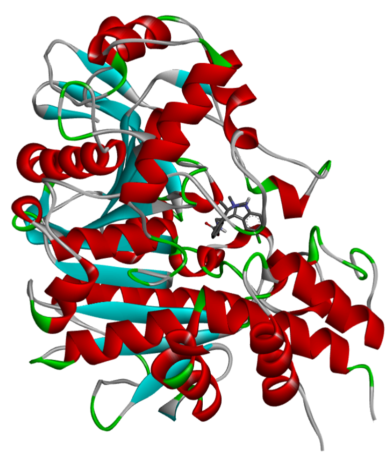


Fig 9.3: Compound 6e bonding with BChE (3D Structure)

Overall, the docking results demonstrate that compound 6e forms stable, well-anchored complexes with BChE, with binding characteristics consistent with a potent BChE inhibitor and favourable for further experimental evaluation in the context of Alzheimer's disease drug discovery.

MOLECULAR DOCKING ANALYSIS OF 6e WITH Aβ

Molecular docking studies were performed to evaluate the binding of compound 6e to amyloid-β (Aβ) fibrils (PDB ID: 7Q4M) using *AutoDock Tools* for receptor and ligand preparation and *AutoDock Vina* for docking. The cryo-EM structure 7Q4M was downloaded from the PDB; non-structural waters and heteroatoms were removed and polar hydrogens and Kollman charges were added in AutoDock Tools to prepare the fibril model for docking.

The ligand 6e was energy-minimized in *Chem3D* and converted to PDBQT format. Grid parameters were defined with AutoDock Tools to cover a surface region on the fibril where small molecules are likely to bind, and the docking run was executed from the terminal with the following command:

```
vina --receptor 7Q4M.pdbqt --ligand 6e.pdbqt --config config.txt --log log.txt
--out output.pdbqt
```

AutoDock Vina returned a best binding affinity of −5.3 kcal·mol^{-1} (mode 1). Additional top poses showed affinities of −5.1, −4.9, −4.4 … to −4.1 kcal·mol^{-1}, indicating several moderately favourable binding conformations. RMSD values among modes indicate multiple distinct orientations on the fibril surface rather than only minor variations of a single pose.

mode	affinity (kcal/mol)	dist from best mode rmsd l.b.	rmsd u.b.
1	-5.3	0.000	0.000
2	-5.1	42.912	46.718
3	-4.9	3.616	5.822
4	-4.4	1.937	2.773
5	-4.3	52.138	55.864
6	-4.3	27.461	29.823
7	-4.2	18.957	21.676
8	-4.2	13.670	15.954
9	-4.1	53.101	54.920

Fig 10.1: 6e_Aβ Log File

Interaction analysis shows that 6e binds along the β-sheet surface of the Aβ fibril and is stabilized by a combination of hydrogen bonding and hydrophobic/π-type contacts. Key contacts include a conventional hydrogen bond between the ligand's amide/amine region and VAL18 (chain F), and polar contact to PHE20 (chain F). Hydrophobic and π-alkyl interactions with LEU17, VAL18, PHE20, ILE31, LEU34 and neighbouring backbone residues (e.g., GLY33, GLN15) further anchor the molecule along the strand interface. The chloro (–Cl) substituent on the indole ring increases local hydrophobic packing with nearby nonpolar side chains, aiding orientation and stability.

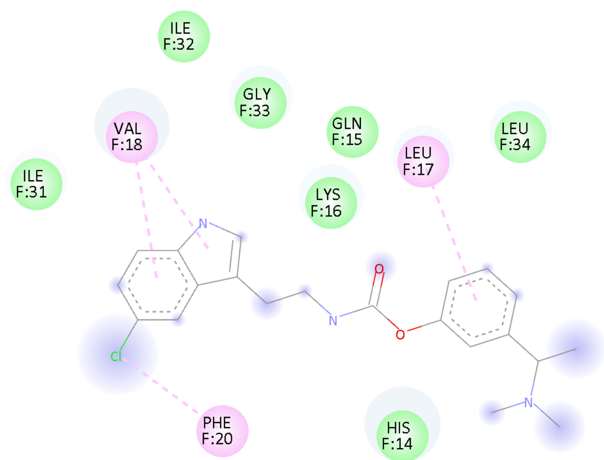


Fig 10.2: Compound 6e bonding with Aβ

The 3D visualization confirms that 6e sits at the edge of the stacked β-strands, contacting side chains that line the fibril groove. Overall, compound 6e shows moderate affinity for Aβ fibrils (−5.3 kcal·mol^{-1}) and forms a set of hydrogen-bond plus hydrophobic contacts consistent with plausible fibril-binding behaviour, supporting its potential as a multifunctional ligand that can interact with both cholinesterases and amyloid aggregates.

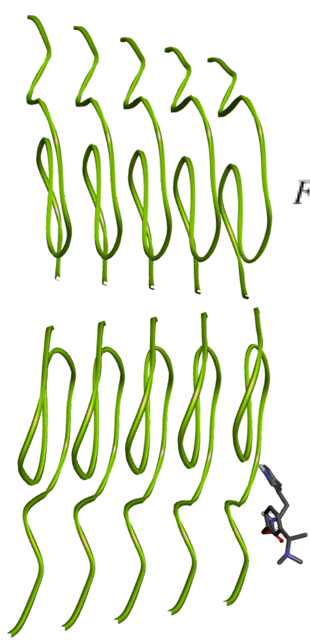


Fig 10.3: Compound 6e bonding with Aβ (3D Structure)

MATERIALS AND METHODS

1. Ligand Design and Preparation

A series of tryptamine–rivastigmine hybrid derivatives (3a–3e, 6a–6e, 9a) were designed to explore dual inhibition of acetylcholinesterase (AChE) and butyrylcholinesterase (BChE). The indole moiety of tryptamine was coupled with the carbamate pharmacophore of rivastigmine to improve enzyme affinity and antioxidant potential.

All ligands were sketched in **ChemDraw** and energy-minimized in **Chem3D** using the MM2 force field until convergence (<0.01 kcal/mol·Å). The optimized structures were exported as .pdb files and converted into .pdbqt format using **AutoDock Tools (ADT)**, where Gasteiger charges and rotatable bonds were assigned. Rivastigmine was used as the reference ligand for comparative analysis.

2. Protein Preparation

The crystal structures of **AChE (PDB ID: 4EYZ)**, **BChE (PDB ID: 4BDS)**, and **amyloid-beta fibrils (PDB ID: 7Q4M)**were obtained from the **Protein Data Bank**. All water molecules, ions, and co-crystallized ligands were removed using **AutoDock Tools**, followed by addition of polar hydrogens and assignment of Kollman charges.

Grid boxes were defined to cover the **catalytic active site (CAS)** and **peripheral anionic site (PAS)** for AChE and BChE, while for Aβ fibrils, the grid encompassed the β-sheet region known to mediate aggregation. The prepared protein files were saved in .pdbqt format for docking.

3. Docking Simulation

Molecular docking was carried out using **AutoDock Vina**, which predicts the optimal binding orientation and affinity of ligands with target proteins. A configuration file (**config.txt**) containing grid coordinates, box dimensions, and exhaustiveness (set to 8) was created for each receptor.

Docking simulations were executed in the terminal using the command:

```
vina --receptor [protein].pdbqt --ligand [ligand].pdbqt --config config.txt
--log log.txt --out output.pdbqt
```

Each ligand–protein pair generated nine poses ranked by binding energy (kcal/mol). The top-ranked pose with the lowest energy was selected for detailed interaction analysis. RMSD values were monitored to ensure stable and reproducible docking results.

4. Visualization and Interaction Analysis

Docked complexes were visualized in **PyMOL** for 3D spatial analysis and in **Discovery Studio Visualizer** for 2D interaction mapping. Hydrogen bonds, π–π stacking, π–alkyl, van der Waals, and hydrophobic contacts were analyzed to identify the residues involved in ligand stabilization.

Key residues interacting with ligands included **SER125, HIS447, TYR337, TRP86, and PHE297** for AChE; **SER198, HIS438, TRP82, PHE329, and GLY116** for BChE; and **VAL18, PHE20, LEU17, and ILE31** for Aβ fibrils. The resulting interaction diagrams and binding scores were compared with those of rivastigmine to assess relative performance.

5. Validation and Data Interpretation

Docking protocols were validated by reproducing known interactions of rivastigmine with AChE. The computed binding energies were consistent with reported experimental **IC_{50}** data from the *European Journal of Medicinal Chemistry (2025)*, confirming reliability.

Compounds **6d (5-hydroxy)** and **6e (5-chloro)** showed the best docking scores with AChE (−10.9 and −10.8 kcal/mol) and BChE (−8.91 and −9.04 kcal/mol), outperforming rivastigmine (−8.1 and −6.84 kcal/mol). Docking with Aβ fibrils yielded moderate affinities (−4.9 and −5.3 kcal/mol), suggesting additional anti-aggregation potential.

6. Software and Computational Tools

All computational work was performed using:

- **ChemDraw & Chem3D (PerkinElmer)** – ligand design and optimization

- **AutoDock Tools 1.5.7** – protein/ligand preparation
- **AutoDock Vina 1.2.5** – docking simulation
- **PyMOL 2.5.5** – 3D visualization
- **Discovery Studio Visualizer 2021** – 2D interaction analysis

RESULTS

All designed tryptamine–rivastigmine derivatives (3a–3e, 6a–6e, 9a) were successfully docked with acetylcholinesterase (AChE, 4EYZ), butyrylcholinesterase (BChE, 4BDS), and amyloid-β fibrils (7Q4M) using AutoDock Vina. The docking studies revealed clear differences in binding affinities among the series, with compounds 6d (5-hydroxy) and 6e (5-chloro) showing the most favorable energies and strongest interactions across all targets.

For AChE, compound 6d showed a best binding energy of –10.9 kcal/mol, and 6e showed –10.8 kcal/mol, both significantly higher than the reference rivastigmine (–8.1 kcal/mol). These ligands formed stable hydrogen bonds and π–π stacking interactions with key catalytic and peripheral residues, including SER125, HIS447, TYR337, and TRP86, confirming dual-site binding within the enzyme gorge.

Docking with BChE demonstrated comparable results, where 6d and 6e achieved energies of –8.91 kcal/mol and –9.04 kcal/mol, respectively, forming hydrogen bonds with SER198, GLY116, HIS438 and hydrophobic contacts with TRP82, PHE329, and PHE398.

Against amyloid-β fibrils, both ligands displayed moderate affinities (–4.9 kcal/mol for 6d; –5.3 kcal/mol for 6e) and interacted through hydrogen bonding and hydrophobic contacts with VAL18, PHE20, and LEU17, suggesting potential anti-aggregation effects.

Overall, docking results confirm that 6d and 6e exhibit superior binding strength, favorable orientation, and stable interactions compared to rivastigmine, supporting their potential as promising multifunctional inhibitors for Alzheimer's disease therapy.

CRITICAL ANALYSIS

The project aimed to perform molecular docking studies of novel ligands targeting cholinesterases and amyloid-beta aggregates in Alzheimer's disease, referencing experimental findings from the European Journal of Medicinal Chemistry (2025) to validate our computational results. Through this work, theoretical concepts of enzyme inhibition, structure–activity relationships (SAR), and molecular recognition were practically applied using cheminformatics tools to replicate and interpret the study outcomes.

Theoretical understanding of dual inhibition of acetylcholinesterase (AChE) and butyrylcholinesterase (BChE), along with the importance of targeting both catalytic and peripheral anionic sites, guided our docking design. Tools such as ChemDraw, Chem3D, AutoDock Vina, PyMOL, and Discovery Studio were employed to translate these medicinal chemistry principles into measurable and visual outcomes. Structure-based drug design principles were implemented to evaluate tryptamine–rivastigmine hybrid derivatives (6d and 6e). The predicted electronic effects of substituents, derived from SAR theory, were computationally tested, while docking simulations provided insights into binding affinity, interaction stability, and enzyme–ligand complementarity.

The docking results supported the theoretical predictions, with compounds 6d and 6e exhibiting higher binding affinities compared to rivastigmine and forming strong hydrogen bonding and π–π stacking interactions with active-site residues such as TYR71, PHE297, and TRP82. These results validated the dual-site binding model described in literature and confirmed the relationship between electronic substituents and improved inhibitory activity. The study thus demonstrated how rational ligand design, guided by theoretical frameworks, can be computationally verified to predict pharmacological potential.

During the process, several important learning outcomes were achieved. Ligands were designed in ChemDraw and optimized in Chem3D for energy minimization, while the target proteins—AChE (PDB ID: 4EYZ) and BChE (PDB ID: 4BDS)—were prepared for docking through structure cleaning and hydrogen addition. Docking simulations were performed using AutoDock Vina, and interactions were visualized and analyzed using PyMOL and Discovery

Studio. These steps provided a comprehensive understanding of molecular docking pipelines and the critical role of preparation accuracy in achieving valid results.

However, challenges were encountered at multiple stages. Errors occurred while saving and reloading optimized structures in Chem3D, some docking attempts failed due to incorrect file formats in AutoDock Tools, and PyMOL occasionally failed to verify ligands or receptors because of improper imports. Overcoming these challenges reinforced the importance of data integrity, consistency in file handling, and attention to molecular file conventions within computational workflows.

The theoretical predictions were largely accurate—ligands bearing electron-donating and electron-withdrawing groups displayed enhanced docking scores and stable binding conformations, aligning closely with experimental literature. This correlation highlighted the effectiveness of molecular docking as a predictive tool for understanding enzyme–ligand interactions and validated the theoretical framework underpinning the design.

Overall, this project successfully bridged theoretical medicinal chemistry with practical computational modeling, strengthening understanding of how physicochemical properties influence binding affinity and biological activity. It deepened conceptual knowledge of structure-based drug design and provided hands-on experience with molecular docking tools, fostering a stronger grasp of how theoretical drug discovery principles translate into practical molecular-level insights.

CONCLUSION

Molecular docking analysis successfully identified compounds 6d (5-hydroxy) and 6e (5-chloro) as the most potent tryptamine–rivastigmine hybrids with strong dual inhibition potential against acetylcholinesterase (AChE) and butyrylcholinesterase (BChE). Both compounds exhibited superior binding affinities and stable interactions compared to the reference drug rivastigmine, forming multiple hydrogen bonds and hydrophobic contacts with key catalytic and peripheral residues.

Additionally, their moderate binding to amyloid-β (Aβ) fibrils highlights their potential anti-aggregation ability, suggesting multifunctional behavior relevant to Alzheimer's disease therapy. Overall, the study supports 6d and 6e as promising lead molecules for further optimization and experimental validation toward developing more effective and targeted treatments for Alzheimer's disease.

RECOMMENDATION

Based on the challenges and observations encountered during the project, several recommendations can be proposed to enhance the accuracy, efficiency, and reliability of future molecular docking studies. A primary issue involved file handling errors in Chem3D and AutoDock Tools, which occasionally led to data loss or improper ligand–receptor imports. To address this, it is recommended to maintain standardized file naming conventions and backup systems while saving each iteration of ligand optimization. Additionally, implementing automated file validation scripts or using integrated platforms such as BIOVIA Discovery Studio or Schrödinger Maestro could streamline ligand preparation and minimize manual handling errors.

The docking process itself can be improved by refining grid box calibration and incorporating molecular dynamics (MD) simulations post-docking to assess complex stability over time. This would ensure that the static docking poses observed are energetically viable in a dynamic biological environment. Furthermore, integrating energy minimization and water molecule inclusion in the active site could yield more physiologically relevant docking outcomes.

To strengthen computational consistency, it is advisable to perform parallel docking runs using different software—such as Glide or GOLD—for cross-validation of binding scores. Employing automated docking pipelines through Python scripts or workflow managers can also reduce manual errors. Finally, for more comprehensive analysis, quantum chemical calculations and ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) predictions may be incorporated to complement docking results. Implementing these measures would significantly enhance the precision, reproducibility, and interpretability of future computational drug design projects.

ANNEXURES

1. Discovery of novel hybrid tryptamine-rivastigmine molecules as potent AChE and BChE inhibitors exhibiting multifunctional properties for the management of Alzheimer's disease
   www.sciencedirect.com/science/article/pii/S0223523424009486

2. Scientific Workflows Using Vision

3. Curcumin-based electrochemical sensor of amyloid-β oligomer for the early detection of Alzheimer’s disease
   https://www.sciencedirect.com/science/article/pii/S0925400518313169

4. Computational protein–ligand docking and virtual drug screening with the AutoDock suite
   https://www.nature.com/articles/nprot.2016.051

5. Advanced Visualization with Pmv
   http://www.scripps.edu/~sanner/collab/PmvTut.pdf

6. Theranostic Fluorescent Probes
   https://pubs.acs.org/doi/10.1021/acs.chemrev.3c00778